

Day 25 — TNPSC Group 2A Study Material

Part A – General Studies: Characteristics of Indian Culture & Unity in Diversity; India as a Secular State & Social Harmony; Prominent Personalities

READING MATERIAL

The final Unit III topic, wrapping up three short clauses: India's cultural characteristics/diversity, its secular character, and notable figures across arts/science/literature/philosophy (several already covered as 'leaders' — this section adds non-political figures).

Q: Real exam: Which of these IS a genuine Fundamental Duty — preserving the rich heritage of composite culture, or safeguarding private property?

A: Preserving the rich heritage of composite culture (Article 51-A(f)) is a genuine Fundamental Duty. 'Safeguarding private property' is NOT — property was removed as a Fundamental RIGHT (now just a legal right, Article 300-A) and was never a Fundamental Duty either.

Q: What does 'Unity in Diversity' mean, in the Indian context?

A: Maintaining a sense of oneness and solidarity DESPITE wide differences in religion, language, culture, caste, region, and lifestyle — the idea that this diversity enriches the nation rather than dividing it.

Q: In what sense is India a 'secular state'?

A: Multiple faiths — Hindu, Muslim, Christian, Sikh, Buddhist, Jain, and others — coexist, with the state not favoring any one religion, allowing each community to practice its own traditions freely.

Q: What is meant by India's 'composite culture', and which historical movements exemplify it?

A: A culture formed through cultural exchange between different communities over centuries — visible in blended architecture, music, and art. The Bhakti and Sufi movements (medieval Hindu devotional and Islamic mystic traditions respectively) both emphasized love, equality, and devotion in ways that transcended strict religious divisions, exemplifying this blending.

Q: Name a few prominent Indian figures in Science (beyond those already covered as freedom-movement leaders).

A: C.V. Raman (Nobel laureate, Raman Effect — already touched in Unit I), Homi J. Bhabha (father of India's nuclear program), and APJ Abdul Kalam (aerospace scientist, later President of India, the 'Missile Man of India').

Q: Name a few prominent Indian figures in Arts (classical dance/music).

A: Rukmini Devi Arundale (revived and popularized Bharatanatyam), and Pandit Ravi Shankar (sitar maestro, who helped bring Indian classical music global recognition).

Q: Name a prominent figure in Philosophy associated with Indian thought.

A: Adi Shankaracharya (8th-century philosopher, systematized Advaita Vedanta) — alongside Swami Vivekananda (already covered under National Renaissance), who brought Vedantic philosophy to a global audience at the 1893 Chicago Parliament of Religions.

MOCK TEST (WITH ANSWERS)

1. Which of the following IS a genuine Fundamental Duty under the Indian Constitution?

- A. To safeguard private property
- B. To preserve the rich heritage of composite culture ✓**
- C. To pay income tax on time
- D. To vote in every election

Explanation: Article 51-A(f) — preserving composite culture. [medium, question-bank-2025]

2. 'Unity in Diversity' in the Indian context primarily refers to:

- A. Forcing cultural uniformity
- B. Maintaining oneness despite religious/linguistic/regional differences ✓**
- C. Eliminating regional languages
- D. A purely economic policy

Explanation: Maintaining solidarity despite wide differences. [easy, web-research]

3. Which medieval movements are commonly cited as exemplifying India's composite culture, emphasizing love and devotion across religious lines?

A. Bhakti and Sufi movements ✓

B. Wahabi and Aligarh movements

C. Brahmo and Arya Samaj movements

D. Justice Party and Self-Respect movements

Explanation: Bhakti (Hindu devotional) and Sufi (Islamic mystic) movements. [medium, web-research]

4. Who is known as the 'Missile Man of India', later becoming President?

A. Homi J. Bhabha

B. C.V. Raman

C. APJ Abdul Kalam ✓

D. Vikram Sarabhai

Explanation: APJ Abdul Kalam. [easy, web-research]

5. Who is credited with reviving and popularizing Bharatanatyam in the 20th century?

A. Rukmini Devi Arundale ✓

B. Pandit Ravi Shankar

C. M.S. Subbulakshmi

D. Balasaraswati

Explanation: Rukmini Devi Arundale. [medium, web-research]

6. Adi Shankaracharya is best known for systematizing which philosophical school?

A. Advaita Vedanta ✓

B. Samkhya

C. Nyaya

D. Mimamsa

Explanation: Advaita Vedanta. [medium, web-research]

Part B – Aptitude & Mental Ability: Volume

READING MATERIAL

Volume is the last of 10 named Aptitude sub-skills. Real exam evidence shows two dominant shapes: 'ratio of dimensions $\rightarrow$ ratio of volumes' (scaling), and 'melt/recast one solid into another shape' (volume is conserved, surface area is not) — both are core techniques worth knowing cold.

Q: Method — ratio of volumes from a ratio of radii (same shape): If the radius of one sphere is half that of another, what is the ratio of their volumes?

A: Volume scales with the CUBE of linear dimensions ($V = \frac{4}{3}\pi r^3$), not directly with the radius. So if $r_1:r_2 = 1:2$, then $V_1:V_2 = 1^3:2^3 = 1:8$.

Q: Method — melting and recasting (volume is conserved, NOT surface area): A sphere of radius 9cm is melted and recast into a cone of the same radius (9cm). Find the cone's height.

A: Set sphere volume = cone volume: $(\frac{4}{3})\pi(9)^3 = (\frac{1}{3})\pi(9)^2h \rightarrow (\frac{4}{3})(729) = (\frac{1}{3})(81)h \rightarrow 972 = 27h \rightarrow h = 36$ cm. The general shortcut: for melting sphere $\rightarrow$ cone of the SAME radius, height = $4 \times$ radius always (since the $(\frac{4}{3})r^3$ vs $(\frac{1}{3})r^2h$ relationship reduces to $h=4r$ when radii match).

Q: Real exam (already seen in an earlier topic, re-noted here): A sphere of diameter 8cm is melted and drawn into a wire of diameter 3mm. Find the wire's length.

A: Volume conserved: $(\frac{4}{3})\pi(4)^3 = \pi(0.15)^2 \times h \rightarrow h \approx 3792.6$ cm ≈ 37.9 m. (Same melt/recast principle as $rO2$, just sphere $\rightarrow$ cylinder/wire instead of sphere $\rightarrow$ cone.)

Q: Method — volume of a cylindrical tank from its capacity: A cylindrical tank has capacity 1,848 m³ and base diameter 14m. Find its depth.

A: $V = \pi r^2 h \rightarrow 1848 = (\frac{22}{7}) \times 7^2 \times h = 154h \rightarrow h = 12$ m.

Q: Method — dug-out soil redistributed into a different shape (volume conserved): A cylindrical well of depth 20m, diameter 14m is dug; the soil is spread evenly to form a cuboid platform of base 20m $\times$ 14m. Find the platform's height.

A: Well volume = $(\frac{22}{7}) \times 7^2 \times 20 = 3,080$ m³. Platform volume = $20 \times 14 \times h = 280h$. Set equal: $280h = 3,080 \rightarrow h = 11$ m.

Q: Formula check: volume of a cube with side 5cm, and the relationship between a cube's volume and surface area.

A: Volume = side³ = $5^3 = 125$ cm³. If a cube's volume is 1000 cm³, its side = $\sqrt[3]{1000} = 10$ cm, so its surface area = $6 \times \text{side}^2 = 6 \times 100 = 600$ cm² (cube surface area = $6 \times \text{side}^2$, from its 6 square faces).

Q: Method — ratio of volumes for cone, sphere, cylinder of equal radius AND equal height: What is that ratio?

A: Cone : Sphere : Cylinder = 1 : 4 : 3, when all three share the same radius r and the same height h (for the sphere, 'height' = its diameter = $2r$ in this comparison). This is a frequently tested ratio worth memorizing directly.

MOCK TEST (WITH ANSWERS)

1. If the radius of sphere A is 3 times the radius of sphere B, what is the ratio of their volumes (A:B)?

- A. 3:1
- B. 9:1
- C. 27:1 ✓
- D. 6:1

Explanation: Volume ratio = (radius ratio)³ = $3^3:1^3 = 27:1$. [easy, authored]

2. A solid sphere of radius 6cm is melted and recast into a cone of the same radius. Find the cone's height.

- A. 18 cm
- B. 24 cm ✓
- C. 12 cm
- D. 6 cm

Explanation: Using the shortcut $h = 4 \times$ radius for sphere $\rightarrow$ same-radius-cone: $h = 4 \times 6 = 24$ cm. [medium, authored]

3. A cylindrical tank has capacity 4,620 m³ and base diameter 14m. Find its depth.

- A. 20 m
- B. 25 m
- C. 30 m ✓

D. 15 m

Explanation: $V = \pi r^2 h \rightarrow 4620 = (22/7) \times 49 \times h = 154h \rightarrow h = 30\text{m}$. [medium, authored]

4. Find the volume of a cube with side 8cm.

A. 512 cm³ ✓

B. 256 cm³

C. 64 cm³

D. 640 cm³

Explanation: $8^3 = 512\text{ cm}^3$. [easy, authored]

5. If a cube's volume is 2,197 cm³, find its surface area.

A. 507 cm²

B. 169 cm²

C. 1014 cm² ✓

D. 338 cm²

Explanation: Side = $\sqrt[3]{2197} = 13\text{cm}$. Surface area = $6 \times 13^2 = 6 \times 169 = 1,014\text{ cm}^2$. [medium, authored]

6. A cone, a sphere, and a cylinder share the same radius and same height. What is the ratio of their volumes (cone:sphere:cylinder)?

A. 1:3:4

B. 1:4:3 ✓

C. 3:4:1

D. 4:3:1

Explanation: Cone:Sphere:Cylinder = 1:4:3 for equal radius and height. [medium, web-research]

Part C – General English: Poem: "The Grumble Family" — L.M. Montgomery

READING MATERIAL

A humorous, cautionary poem about a family that complains about absolutely everything, living on 'Complaining Street'. Real coaching-site evidence tests the poem's setting, irony, and a couple of figures of speech.

Q: Who wrote "The Grumble Family", and what is its central theme?

A: L.M. Montgomery. Central theme: the danger of constant complaining — the poem warns against habitual negativity and urges readers to adopt a positive attitude instead.

Q: Where does the Grumble family live?

A: On 'Complaining Street', in the city of 'Never-Are-Satisfied', beside the 'River of Discontent' — all symbolically named to match the family's perpetually negative outlook.

Q: What is ironic about the Grumble family's behaviour?

A: They complain even when things are perfect — rain or sun, summer or winter, high or humble status, nothing satisfies them, which is the central irony: their unhappiness isn't caused by genuinely bad circumstances, but by their own habit of complaining regardless of circumstances.

Q: Identify the figure of speech: "The River of Discontent beside."

A: Imagery — a vivid descriptive image (a symbolic river) representing the family's emotional state.

Q: Identify the figure of speech: "Growl at the rain and growl at the sun."

A: Alliteration — repeated initial sounds, here also reflected in similar examples like 'They'd growl that they'd nothing to grumble about'.

Q: What does the poet ultimately want readers to do, per the poem's message?

A: Stop complaining and develop a positive attitude — to 'walk with a smile and a song' rather than grumble regardless of circumstances.

MOCK TEST (WITH ANSWERS)

1. "The Grumble Family" was written by:

- A. Rudyard Kipling
- B. L.M. Montgomery ✓**
- C. Jane Taylor
- D. R.L. Stevenson

Explanation: L.M. Montgomery. [easy, web-research]

2. The main theme of the poem is:

- A. City life
- B. The danger of complaining ✓**
- C. Weather conditions
- D. Family reunion

Explanation: The danger of complaining. [easy, web-research]

3. What is the name of the street the Grumble family lives on?

- A. Lonely Street
- B. Happy Lane
- C. Complaining Street ✓**
- D. River Road

Explanation: Complaining Street. [easy, web-research]

4. What is ironic about the Grumble family's behaviour?

- A. They never complain
- B. They are praised for kindness
- C. They complain even when things are perfect ✓**
- D. They're always joyful

Explanation: They complain regardless of how good circumstances actually are. [medium, web-research]

5. "The River of Discontent beside" is an example of which device?

- A. Simile
- B. Metaphor
- C. Alliteration

D. Imagery ✓

Explanation: Imagery. [medium, web-research]

6. "Growl at the rain and growl at the sun" uses which figure of speech?

- A. Metaphor
- B. Personification
- C. Hyperbole

D. Alliteration ✓

Explanation: Alliteration. [medium, web-research]